

Notes: Dang, Gorton, Holmström, and Ordoñez (AER, 2017)

Banks as Secret Keepers

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1 Big picture

- **Question.** Why do banks exist, and why are they naturally opaque, if capital markets are better at aggregating and revealing information?
- **Core answer.** Banks produce money-like short-term debt. To make bank liabilities trade at par, banks must keep the value of their backing assets information-insensitive.
- **Main distinction.** Capital markets reveal information through prices. This creates **risky liquidity**. Banks hide information and create **safe liquidity**.
- **Bank secrecy is productive.** Opacity prevents privately informed agents from acquiring or using detailed information about bank assets. This keeps all traders symmetrically uninformed and makes bank liabilities liquid.
- **The trade-off.** Information revelation helps allocate investment efficiently, but it also makes claims state-contingent and less money-like. Secrecy preserves money-like liquidity, but can require banks to scale down lending or reduce the amount of safe liquidity they provide.
- **Capital market outcome.** Markets reveal the project state at the interim date. The early investor can sell shares, but the price is state-contingent. In bad states, the investor cannot fully meet liquidity needs.
- **Bank outcome without information acquisition.** A bank keeps the file secret. Deposits are paid at a fixed amount at the interim date, so the early investor obtains safe liquidity and the first-best allocation can be implemented.
- **Bank outcome with costly private information acquisition.** If late investors can pay a cost to learn the hidden state, the bank must discourage information production. It can do so by reducing investment or reducing money provision.
- **Sorting result.** Banks dominate for assets that are sufficiently opaque, safe, or hard to evaluate. Capital markets dominate when information is cheap to produce or assets are too transparent/risky for banks to preserve opacity cheaply.
- **Economic takeaway.** Banks are not merely information producers. They are also **secret keepers**. Their opacity is not an accidental defect; it is central to producing private money.

2 Model environment and measurement

2.1 Agents, dates, and goods

- There is one good and three dates: $t \in \{0, 1, 2\}$.
- There is a firm F , one early consumer E , and $N > 1$ identical late consumers L .
- The early consumer is born at date 0 with endowment e .
- Late consumers are born at date 1 with endowment e .
- Both consumers have urgent liquidity needs in the period after they are born.

Preferences can be summarized as

$$U_F = \sum_{t=0}^2 c_{F,t},$$

$$U_E = \sum_{t=0}^2 c_{E,t} + \alpha \min\{c_{E,1}, k\},$$

$$U_L = \sum_{t=1}^2 c_{L,t} + \alpha \min\{c_{L,2}, k\},$$

where $\alpha > 1$ measures the extra value of urgent consumption and k is the satiation point of the liquidity need.

Interpretation: The kinked utility function makes the consumer risk-averse over whether the liquidity need is met, while keeping the model simple once consumption exceeds k .

2.2 Projects

The firm has two projects:

- a **lemon**, which never produces output;
- a **worthy project**, which costs w at date 0 and pays $x > w$ at date 2 in the good state g with probability λ , and zero in the bad state b with probability $1 - \lambda$.

The worthy project is linearly divisible. If the firm operates fraction η of the project, investment costs ηw and pays ηx in the good state.

The key assumptions are

$$\begin{aligned} \lambda x &> w, \\ e &> k, \quad e > w, \quad e < k + w, \\ 2e &> 2k + w. \end{aligned}$$

Interpretation: The project is socially valuable, but the early consumer cannot both finance the project and fully self-insure his date-1 liquidity need. This creates the motive for financial intermediation.

2.3 Information structure

- A project file identifies the worthy project at date 0.
- Both banks and market agents can use the file to distinguish the worthy project from the lemon.
- Late consumers have superior expertise: if they see the file at date 1, they can determine whether the worthy project will succeed or fail.
- Banks can keep the file secret.
- Markets cannot keep the file secret.
- Markets involve simultaneous interaction among late consumers; banks interact with one randomly selected late consumer.

Interpretation: The same information has two roles. At date 0, it is useful because it identifies good projects. At date 1, it is dangerous for liquidity because it makes claims state-contingent.

2.4 Autarky and first best

Let

$$z = e - w.$$

If the early consumer finances the project at date 0, he has only $z < k$ left for date-1 liquidity.

In autarky, consumers store their own endowments and the project is not financed:

$$\mathbb{E}U_E^A = \mathbb{E}U_L^A = e + \alpha k.$$

In the first best:

- the worthy project is fully financed at date 0;

- late consumers transfer $k - z$ to the early consumer at date 1;
- the early consumer consumes k at date 1 regardless of the project state;
- the firm receives the social surplus:

$$\mathbb{E}U_F^{\text{FB}} = \lambda x - w.$$

Interpretation: The first best requires ex ante risk sharing between early and late consumers. The model asks whether capital markets or banks can implement this risk sharing.

3 Models and theoretical specifications

3.1 Capital markets: information revelation and risky liquidity

At date 0, the firm issues a security that pays $s^M(y)$ at date 2, where $y \in \{b, g\}$. In a fully revealing capital market, late consumers know the state at date 1. Therefore:

$$s^M(b) = 0.$$

The early consumer finances the security only if his expected utility is at least autarky. His utility from financing is

$$U_E^{\text{Finance}} = (1 + \alpha)z + \lambda [s^M(g) + \alpha(k - z)].$$

The equilibrium good-state payment is

$$s^M(g) = \min \left\{ \frac{w}{\lambda} + \frac{\alpha(1 - \lambda)}{\lambda}(k - z), x \right\}.$$

The welfare loss relative to first best is

$$\mathbb{E}U_F^{\text{FB}} - \mathbb{E}U_F^M = \lambda s^M(g) - w = \min\{\alpha(1 - \lambda)(k - z), \lambda x - w\}.$$

Interpretation: Capital markets produce liquidity because shares can be sold. But the shares sell at state-contingent prices, so the early consumer bears liquidity risk in the bad state and must be compensated by a risk premium.

3.2 Banking without private information acquisition: secret keeping and safe liquidity

A bank observes the file at date 0 and can keep it secret at date 1. The bank receives deposits and writes contracts with both consumers.

A first-best implementation sets

$$r_1^E = k, \quad r_2^E(b) = 0, \quad s^B(b) = 0.$$

In the bad state, the bank's assets at date 2 are

$$A_b = e + z - k.$$

The early consumer's break-even condition implies

$$r_2^E(g) = \frac{e - k}{\lambda}.$$

The late consumer can be promised

$$r_2^L(b) = A_b = e + z - k > k.$$

The firm sells the bank a claim with

$$s^B(g) = \frac{w}{\lambda}.$$

The firm's expected surplus is

$$\lambda x - \lambda s^B(g) = \lambda x - w.$$

Interpretation: The bank can implement the first best because it does not reveal the file at date 1. The early consumer's date-1 payment is non-contingent, like a deposit. This is safe liquidity.

3.3 Private information acquisition

Now suppose the late consumer can pay information cost γ to learn the hidden state from the file. If she learns that the state is bad, she refuses to deposit.

The bank must satisfy the no-information-production constraint:

$$r_2^L(b) \geq e - \frac{\gamma}{1 - \lambda}.$$

Using the first-best banking contract $r_2^L(b) = e + z - k$, this becomes

$$(1 - \lambda)(k - z) \leq \gamma.$$

Define the excess incentive to acquire information:

$$\Delta \equiv k - z - \frac{\gamma}{1 - \lambda}.$$

If $\Delta \leq 0$, the bank can implement the first best. If $\Delta > 0$, secrecy is fragile and the bank must distort its contract.

Interpretation: The value of becoming informed is the avoided loss in the bad state, $(1 - \lambda)(k - z)$. Banks can keep secrets only when this private value is below the cost of information production.

3.4 Bank distortion 1: scale down investment

If $\Delta > 0$, the bank can reduce investment and store safe assets. This raises resources available in the bad state and makes information acquisition less attractive.

The maximum investment scale consistent with no information acquisition is

$$\eta^B = 1 - \frac{\Delta}{w}.$$

The welfare loss relative to first best is

$$(1 - \eta^B)(\lambda x - w) = \Delta \left(\frac{\lambda x}{w} - 1 \right).$$

Interpretation: Scaling down investment preserves the early consumer's safe liquidity payment k , but reduces the amount of productive investment.

3.5 Bank distortion 2: reduce money provision

Alternatively, the bank can reduce the early consumer's safe date-1 payment:

$$r_1^E = k - \Delta.$$

To keep the early consumer willing to deposit, the bank compensates him with more good-state date-2 consumption. The welfare loss relative to first best is

$$\alpha \Delta.$$

Interpretation: Reducing money provision keeps investment at full scale but makes the early consumer bear liquidity shortfall at date 1.

3.6 Which bank distortion is cheaper?

The bank compares the cost of investment distortion,

$$\Delta \left(\frac{\lambda x}{w} - 1 \right),$$

with the cost of money-provision distortion,

$$\alpha \Delta.$$

Therefore, the bank distorts money provision rather than investment iff

$$\frac{\lambda x}{w} - 1 > \alpha.$$

Interpretation: When the project has a high expected net return, cutting investment is costly, so the bank instead reduces money-like liquidity. When liquidity is very valuable relative to project return, the bank preserves liquidity and scales down investment.

3.7 Capital markets with partial investment

Capital markets can also choose not to fully fund the project. If full funding requires too much compensation for liquidity risk, the market funds only the fraction

$$\eta^M = \frac{e - k}{w}.$$

The capital-market welfare loss is

$$\Omega = \min \left\{ \alpha(1 - \lambda), \frac{\lambda x}{w} - 1 \right\} (k - z).$$

Interpretation: Markets either fund the project fully and create risky liquidity, or scale down investment so the early consumer can self-insure his liquidity need.

3.8 Banks versus markets

If $\Delta \leq 0$, banks dominate because they implement the first best.

If $\Delta > 0$, banks incur the cheaper of the two distortion costs:

$$L^B = \min \left\{ \alpha, \frac{\lambda x}{w} - 1 \right\} \Delta.$$

Capital markets dominate when

$$\Omega < L^B.$$

Interpretation: The institution that funds the firm is the one that creates liquidity at the lower welfare cost. Banks win when secrecy is cheap. Markets win when keeping secrets requires too much distortion.

4 Mechanism and theoretical design

4.1 Why capital markets cannot produce bank money

- Capital markets aggregate information through competition among expert traders.
- This informational efficiency makes prices reveal the state.
- Once prices reveal the state, claims cannot maintain a stable value across states.
- The early consumer can trade the claim, but the amount of liquidity obtained is risky.

Interpretation: Price discovery is useful for allocation but harmful for producing par-like money.

4.2 Why banks can produce money-like liabilities

- Banks can keep loan details secret.
- Secret keeping prevents late consumers from learning the state.
- Without state information, the late consumer deposits at the bank.
- The early consumer receives a non-contingent date-1 withdrawal.

Interpretation: Bank opacity creates symmetric ignorance. This is exactly what makes bank liabilities liquid.

4.3 Why secrecy can be costly

- If information is cheap to acquire, the late consumer may privately learn the state.
- To stop this, the bank must make the bad-state payoff high enough that learning is not useful.
- Raising the bad-state payoff requires either storing safe assets or reducing promised liquidity.
- Both actions create welfare costs.

Interpretation: The model does not say banks should always hide information. It says banks hide information when doing so is cheaper than letting markets reveal it.

4.4 What determines borrower sorting

Borrowers sort into banks or markets based on:

- λ : probability the project succeeds;
- x/w : expected project return per unit of investment cost;
- γ : cost of producing private information;
- $k - z$: size of the liquidity shortfall that bank money must cover;
- α : marginal value of meeting liquidity needs.

Interpretation: Young, small, opaque, and difficult-to-evaluate firms are natural bank borrowers. More transparent firms, or firms whose information is cheap to produce, are more natural capital-market borrowers.

4.5 Limits of the theory

- The model is highly stylized: three dates, one firm, one project, and simplified preferences.
- It does not model bank runs, deposit insurance, banker moral hazard, or regulatory supervision directly.
- It abstracts from many details of bank asset screening and monitoring.
- It treats secrecy as a technology of banks and price discovery as a technology of markets.
- The goal is not empirical estimation, but a theory of why opacity can be socially useful.

Interpretation: The paper's main force is conceptual. It gives a reason why transparency can be bad for money-like claims even if it is good for information aggregation.

5 Tables and figures

5.1 Table 1 + Figure 1: Capital markets create risky liquidity

TABLE 1—TIMING: MODEL WITH CAPITAL MARKETS	
Date $t = 0$	The firm goes to the market agent, who identifies the worthy project from the firm's files. The firm raises w from the market agent by issuing a security that pays $s^M(b)$ in case of failure and $s^M(g)$ in case of success. The market agent obtains w by selling the security to E and making files public.
Date $t = 1$	All L observe the file and learn the state, b or g. E sells a fraction of his security in the market to L.
Date $t = 2$	Project payoffs are realized. The firm pays the security.

(a) Table 1: timing with capital markets.

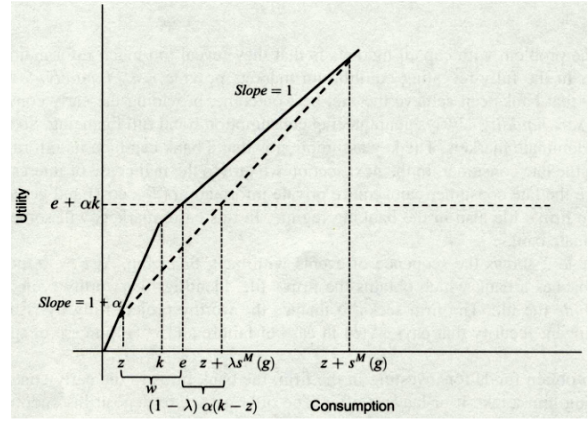


FIGURE 1. COSTS OF CAPITAL MARKETS

(b) Figure 1: cost of risky liquidity in capital markets.

Read:

- At date 0, the firm goes to the market agent, who identifies the worthy project and makes the file public.
- The firm issues a state-contingent security.
- At date 1, all late consumers observe the file and learn whether the project will succeed or fail.
- The early consumer sells part of the security in the market.
- Figure 1 shows that in the bad state the early consumer consumes only $z < k$.
- The early consumer therefore demands a risk premium equal to the liquidity-risk wedge.

Economic message: Capital markets provide liquidity through trading, but because prices reveal information, the liquidity is risky. The claim cannot trade like money.

5.2 Table 2 + Figure 2: Banks keep secrets, but secrecy may require distortion

TABLE 2—TIMING: MODEL WITH FINANCIAL INTERMEDIARIES	
Date $t = 0$	E deposits e with the bank. The bank promises a non-contingent payment r_1^E at $t = 1$ and a conditional payment $r_2^E(g)$ if the worthy project succeeds and $r_2^E(b)$ if it fails at $t = 2$. The firm goes to the bank, who identifies the worthy project from the firm's files. The bank lends w to the firm with a loan contract, where the firm pays $s^b(g)$ if the project fails and $s^g(g)$ if it succeeds. The bank chooses whether to keep the firm's files secret.
Date $t = 1$	L deposits e and the bank promises a contingent payment $r_2^L(g)$ if the project succeeds and $r_2^L(b)$ if the project fails at $t = 2$. E withdraws r_1^E and consumes.
Date $t = 2$	Project payoffs are realized. The firm pays the security and the bank pays to the consumers.

(a) Table 2: timing with financial intermediaries.

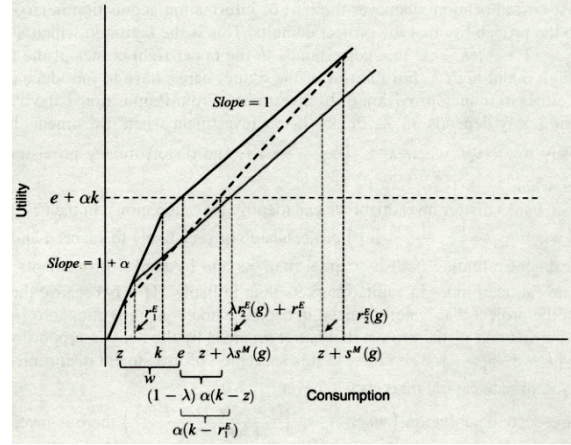


FIGURE 2. CASE IN WHICH CAPITAL MARKETS DOMINATE BANKS

(b) Figure 2: case where markets dominate banks.

Read:

- At date 0, the early consumer deposits with the bank.
- The bank identifies the worthy project and decides whether to keep the file secret.
- The bank lends to the firm and promises the early consumer a non-contingent date-1 payment.
- At date 1, a late consumer deposits with the bank if information acquisition is not attractive.
- Figure 2 compares the utility cost of bank money-provision distortion with the cost of market liquidity risk.
- Markets dominate when the distortion needed to keep bank information secret is larger than the risk premium created by markets.

Economic message: Banks can make liquidity safe only by preserving opacity. If preserving opacity is too costly, markets become the cheaper source of funding.

5.3 Figure 3: Regions of financing

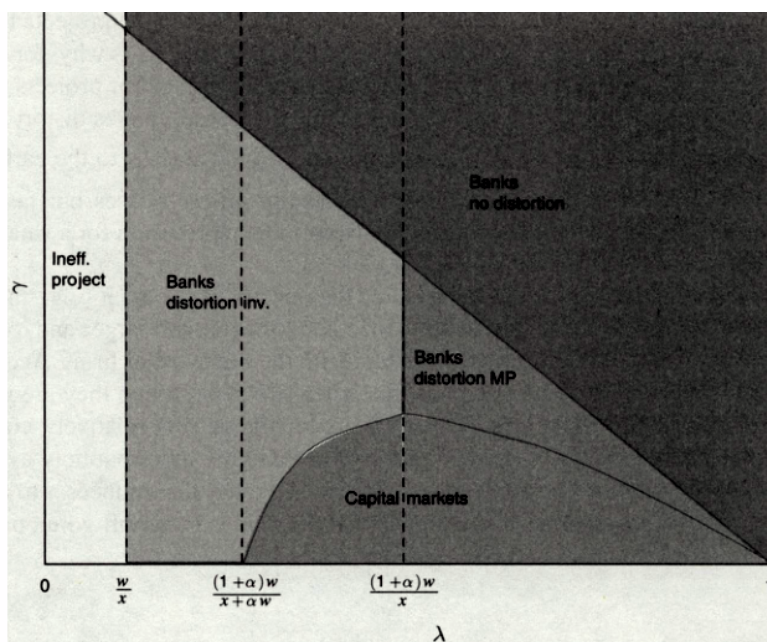


FIGURE 3. REGIONS OF FINANCING

Figure 3: Financing regions as a function of success probability and information cost.

Read:

- The horizontal axis is the success probability λ .
- The vertical axis is information-acquisition cost γ .
- Low λ projects can be inefficient and may not be financed.
- High γ and high λ generate the “banks no distortion” region.
- Intermediate regions require banks to distort either investment or money provision.
- The capital-market region appears where information is cheap enough, or the bank distortion is large enough, that price discovery is cheaper than secrecy.

Economic message: The figure summarizes the paper’s sorting logic: opaque/safe projects belong in banks; transparent or information-sensitive projects migrate to capital markets.

6 Short summary

- Banks produce short-term debt that functions as money.
- Money-like debt must be information-insensitive: it should trade at par without costly valuation.
- Capital markets reveal information through prices, so they produce state-contingent and risky liquidity.
- Banks can keep information about backing assets secret, preserving symmetric ignorance and safe liquidity.

- Without private information acquisition, banks implement the first best and dominate capital markets.
- If late consumers can acquire hidden information cheaply, banks must discourage information production.
- Banks can discourage information acquisition by reducing investment or reducing safe liquidity.
- The cheaper distortion depends on the value of investment relative to the value of liquidity.
- Capital markets dominate when the cost of bank secrecy exceeds the cost of market liquidity risk.
- The model explains why banks naturally hold opaque assets and why transparency can undermine private money creation.

7 Conclusion

7.1 Core conclusion

Dang, Gorton, Holmström, and Ordoñez argue that banks are valuable because they keep secrets. Banks do not merely screen borrowers; they preserve opacity so that their liabilities can remain money-like. Capital markets, by contrast, are designed to reveal information. This informational efficiency makes markets good at price discovery but bad at producing stable-value money.

7.2 Economic intuition

The intuition is that information can destroy liquidity. If everyone knows a claim is bad, it trades at a discount; if some traders may know more than others, uninformed traders fear adverse selection. Bank secrecy prevents this by keeping detailed asset information out of trade. But secrecy is not free: when investors can cheaply produce information, the bank must either store more safe assets or provide less money-like liquidity.

7.3 Takeaway

The paper reframes bank opacity as a feature rather than a bug. Opaque banks and transparent markets are complementary institutions. Banks are best for assets that can be kept information-insensitive; markets are best when information should be revealed and priced.